Zero-Sum Games and Bayesian Games

Call or Fold ?

SciMathSoc

July 8, 2026

Outline

Why Study Zero-Sum Games?

Many real-world situations involve strategic conflict where one player's gain is exactly another player's loss.

Examples include:

- Chess
- Poker
- Cybersecurity
- Military planning
- Online auctions

Central Question

How should a rational player choose a strategy when facing an intelligent opponent?

Zero-Sum Games

Definition

A zero-sum game is a game in which the sum of the payoffs of all players is always zero.

If Player 1 receives payoff u_1
and Player 2 receives payoff u_2
then $u_1 + u_2 = 0$.
Hence,

$$u_2 = -u_1.$$

Examples of Zero-Sum Games

- Chess
- Rock–Paper–Scissors
- Matching Pennies
- Poker (Heads-up)
- Military attack–defense

Winner	Loser
$+1$	-1
$+5$	-5
$+$x	-x

Why Only One Payoff Matrix?

Suppose Player 1 has payoff matrix

$$A = \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$$

Then Player 2 automatically has

$$-A = \begin{bmatrix} -2 & 1 \\ 3 & -4 \end{bmatrix}$$

Only one payoff matrix is sufficient to completely describe a zero-sum game.

Components of a Zero-Sum Game

Every zero-sum game consists of four basic components.

Component	Description
Players	Decision makers in the game
Strategies	Available actions for each player
Payoffs	Rewards or losses after each action
Rules	Order and information structure

Example

In Rock–Paper–Scissors:

Players = 2

Strategies = Rock, Paper, Scissors

Payoff = Win, Lose or Draw

Pure Strategies

Definition

A **pure strategy** is a deterministic plan in which a player always chooses the same action whenever a decision is required.

Examples:

- Always play Rock.
- Always Defend.
- Always Bid High.
- Always Fold.

Pure strategies involve **no randomness**.

Mixed Strategies

Definition

A **mixed strategy** assigns probabilities to each pure strategy.

Example:

Strategy	R	P	S
Probability	0.4	0.3	0.3

The player randomizes according to these probabilities.

Mixed strategies prevent opponents from exploiting predictable behavior.

Pure vs Mixed Strategies

Pure Strategy	Mixed Strategy
Single action	Probability distribution
Deterministic	Randomized
Easy to predict	Difficult to predict
May be exploitable	Often optimal

Many zero-sum games have no optimal pure strategy.
Mixed strategies overcome this limitation.

Example: Matching Pennies

Two players simultaneously reveal a coin.

- If both coins match, Player 1 wins.
- Otherwise, Player 2 wins.

	H	T
H	1	-1
T	-1	1

Question:
Should either player always choose Heads?

Why Do We Need Mixed Strategies?

Suppose Player 1 always chooses Heads.

Player 2 quickly learns this and always chooses Tails.

Player 1 now loses every game.

Key Idea

Being predictable allows your opponent to exploit you.

Randomization removes predictability.

Strategy Space

Suppose a player has n pure strategies.

A mixed strategy is represented by

$$p = (p_1, p_2, \ldots, p_n)$$

where

$$p_i \geq 0, \qquad \sum_{i=1}^{n} p_i = 1.$$

Each p_i denotes the probability of choosing strategy i.

Expected Payoff

Suppose Player 1 chooses

$$p = (p_1, \dots, p_m)$$

and Player 2 chooses

$$q = (q_1, \dots, q_n).$$

If A is the payoff matrix, the expected payoff is

$$E = p^T A q.$$

This quantity measures the average payoff when both players randomize.

Towards Optimal Strategies

A rational player wishes to answer:

- Which strategy guarantees the best outcome?
- Can I protect myself against every opponent?
- Is there a strategy that is always optimal?

These questions lead naturally to the concepts of

Maximin and **Minimax**.

Maximin Strategy

Player 1 wants to be prepared for the **worst-case scenario**.

For each strategy:

- Find the minimum payoff.
- Choose the strategy with the largest minimum payoff.

$$\boxed{\text{Maximin} = \max_i \left(\min_j a_{ij} \right)}$$

Idea

"Choose the strategy that guarantees the best worst-case payoff."

Example: Finding the Maximin

Player 1's payoff matrix

	B_1	B_2
A_1	2	-1
A_2	4	1

Row minima:

$$A_1 : -1 \qquad A_2 : 1$$

Therefore,

$$\boxed{\text{Maximin} = 1}$$

Player 1 chooses strategy A_2.

Minimax Strategy

Player 2 wants to keep Player 1's payoff as small as possible.

For each column:

- Find the maximum payoff.
- Choose the column with the smallest maximum.

$$\boxed{\text{Minimax} = \min_j \left(\max_i a_{ij} \right)}$$

Idea

"Minimize your opponent's best possible payoff."

Example: Finding the Minimax

Player 1's payoff matrix

	B_1	B_2
A_1	2	-1
A_2	4	1

Column maxima:

$$B_1 : 4 \qquad B_2 : 1$$

Therefore,

$$\boxed{\text{Minimax} = 1}$$

Player 2 chooses strategy B_2.

Saddle Point

If

$$\boxed{\text{Maximin} = \text{Minimax}}$$

then the game has a **saddle point**.

At a saddle point,

- Neither player benefits by changing strategy.
- Both players have an optimal pure strategy.

The common value is called the **value of the game**.

Example: Saddle Point

	B_1	B_2
A_1	2	-1
A_2	4	$\boxed{1}$

Row minima:

$$-1,\ 1$$

Column maxima:

$$4,\ 1$$

Since

$$\boxed{\text{Maximin} = \text{Minimax} = 1}$$

the entry 1 is the saddle point.

When There Is No Saddle Point

Consider

	B_1	B_2
A_1	2	5
A_2	4	1

Maximin

$$= \max(2, 1) = 2$$

Minimax

$$= \min(4, 5) = 4$$

Since

$$2 \neq 4,$$

What If There Is No Saddle Point?

Without a saddle point,

- No optimal pure strategy exists.
- Players can exploit predictable behaviour.
- Randomization becomes necessary.

Next Topic

Mixed strategies allow players to achieve an optimal expected payoff even when no saddle point exists.

Mixed Strategies

When no saddle point exists, players should avoid being predictable.

Instead of always choosing the same action, a player chooses each strategy with a certain probability.

$$P(A_1) = p, \qquad P(A_2) = 1 - p$$

Similarly,

$$P(B_1) = q, \qquad P(B_2) = 1 - q.$$

Goal

Find the probabilities that maximize the expected payoff.

Example: A 2×2 Game

Consider the payoff matrix for Player 1:

	B_1	B_2
A_1	4	2
A_2	1	5

Observe that

$$\text{Maximin} = 2, \qquad \text{Minimax} = 4.$$

Since they are different, there is no saddle point.

Finding the Optimal Mixed Strategy

Suppose Player 1 plays

$$A_1 \text{ with probability } p,$$

and

$$A_2 \text{ with probability } 1 - p.$$

Player 1 wants to choose p so that Player 2 is indifferent between both columns.

This makes Player 2 unable to exploit any predictable behaviour.

Expected Payoff Against Each Column

If Player 2 chooses B_1,

$$E(B_1) = 4p + 1(1 - p)$$

$$= 3p + 1$$

If Player 2 chooses B_2,

$$E(B_2) = 2p + 5(1 - p)$$

$$= 5 - 3p$$

The Indifference Principle

Player 1 chooses p such that $\mathrm{E}(\mathrm{B}_1) = E(B_2)$.

Therefore,

$$3p + 1 = 5 - 3p.$$

Solving,

$$6p = 4$$

$$\boxed{p = \frac{2}{3}}$$

Player 1 should play

$$(A_1, A_2) = \left(\frac{2}{3}, \frac{1}{3}\right).$$

Value of the Game

Substituting

$$p = \frac{2}{3}$$

into either expected payoff,

$$V = 3\left(\frac{2}{3}\right) + 1$$

$$V = 3.$$

Definition

The expected payoff obtained under optimal play is called the **value of the game**.

Finding Player 2's Strategy

Similarly, let Player 2 play

$$(B_1, B_2) = (q, 1 - q).$$

Player 2 chooses q so that Player 1 is indifferent between both rows.

The same indifference principle is applied.

This gives the optimal mixed strategy for Player 2.

General Procedure

To solve a 2×2 zero-sum game:

1. Check for a saddle point.
2. If one exists, use pure strategies.
3. Otherwise, assume mixed strategies.
4. Compute expected payoffs.
5. Apply the indifference principle.
6. Obtain the optimal probabilities.
7. Compute the value of the game.

von Neumann's Minimax Theorem

Theorem

Every finite two-player zero-sum game possesses an optimal mixed strategy. Moreover,

$$\boxed{\max\min = \min\max}$$

when mixed strategies are allowed.

This remarkable result was proved by John von Neumann in 1928 and forms the foundation of modern game theory.

Key Takeaways

- Zero-sum games model situations of complete conflict.
- Players seek to maximize their own payoff while minimizing their opponent's payoff.
- Pure strategies are optimal only when a saddle point exists.
- Mixed strategies introduce randomness to avoid exploitation.
- The value of the game is the expected payoff under optimal play.
- The Minimax Theorem guarantees optimal mixed strategies for finite zero-sum games.

From Zero-Sum Games to Bayesian Games

So far, we have assumed that:

- All players know the rules of the game.
- Every player knows the available strategies.
- Every player knows the payoff matrix.

These are called **games of complete information**.

Question

What happens if players have private information?

Examples include auctions, poker, negotiations, and online markets.

Strategies in Extensive Games

Before studying Bayesian games, we introduce two important concepts.

A player may choose actions in two different ways:

1. Mixed Strategies
2. Behavioral Strategies

Although they appear similar, they randomize at different stages of the game.

Mixed Strategy

Definition

A mixed strategy is a probability distribution over complete pure strategies.

The player chooses one pure strategy randomly before the game begins.

Example:

$$P(S_1) = 0.6, \qquad P(S_2) = 0.4.$$

Once selected, the chosen strategy is followed throughout the game.

Behavioral Strategy

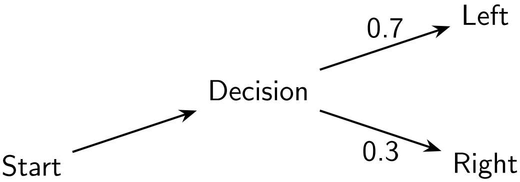
Definition

A behavioral strategy specifies a probability distribution over actions at every decision point.

Instead of randomizing once,
the player randomizes whenever a decision is reached.

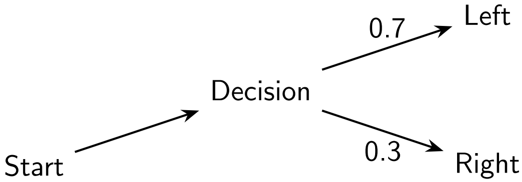
Different information sets may use different probabilities.

Behavioral Strategy Example



At this information set,
the player chooses
Left with probability 0.7
and
Right with probability 0.3.

Behavioral Strategy Example



At this information set,
the player chooses
Left with probability 0.7
and
Right with probability 0.3.

Mixed vs Behavioral Strategies

Mixed Strategy	Behavioral Strategy
Randomize over pure strategies	Randomize at each decision point
Choice made once	Choice made repeatedly
Before the game starts	During the game
Probability on strategies	Probability on actions

Mixed vs Behavioral Strategies

Question:
Do mixed and behavioral strategies always produce the same outcomes?

Answer:

- For games with **perfect recall**, they are equivalent.
- This is known as **Kuhn's Theorem**.

Therefore,
either representation may be used in most extensive-form games.

Perfect Recall

A player has perfect recall if they never forget

- what they previously knew,
- what actions they have taken.

Most extensive-form games satisfy this property.

Under perfect recall,
mixed strategies

$$\Longleftrightarrow$$

behavioral strategies.

Motivation for Bayesian Games

Many real-world games involve uncertainty about other players.

Examples:

- Poker
- First-price auctions
- Job market signaling
- Security games
- Negotiation

Players know the rules,
but they do not know each other's private information.

From Complete to Incomplete Information

Zero-Sum Games

$$\Downarrow$$

Complete Information

Bayesian Games

$$\Downarrow$$

Incomplete Information

Next, we introduce the notion of

$$\boxed{\textbf{Types}}$$

which forms the foundation of Bayesian games.

Incomplete Information

In many strategic situations, players do not possess complete information.

A player may be uncertain about

- another player's payoff,
- preferences,
- cost,
- valuation,
- or available information.

Examples

Auction bidders do not know others' valuations.
Poker players do not know opponents' cards.

Complete vs Incomplete Information

Complete Information	Incomplete Information
All payoffs known	Some payoffs unknown
Strategies known	Private information exists
No uncertainty about players	Uncertainty about player types
Classical Games	Bayesian Games

Private Information

Suppose two firms compete in a market.

Firm A knows its production cost.
Firm B does not.

Possible costs:

High Cost   Low Cost

Firm A's cost is called its

$\boxed{\textbf{Private Information}}$

Nature

Bayesian games introduce an additional player called

$\boxed{\textbf{Nature}}$

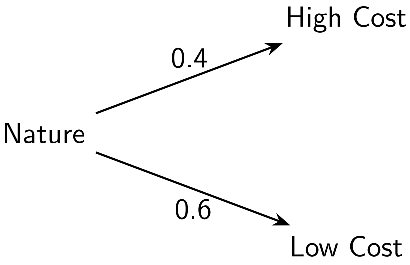
Nature randomly selects the type of each player before the game begins.

Players know

- their own type,
- the probability distribution over all types.

They do **not** know the types of other players.

Nature Chooses Player Types



Nature selects player types according to known probabilities.

Player Types

Definition

A player's type summarizes all private information relevant to decision making.

A type may represent

- valuation,
- production cost,
- skill level,
- risk preference,
- private signal.

Different types may choose different strategies.

Example

Suppose a bidder participates in an auction.

Nature chooses

$$v = \begin{cases} 100, & 0.3 \\ 200, & 0.5 \\ 300, & 0.2 \end{cases}$$

The bidder observes v but other bidders do not.

The valuation v is the bidder's type.

Beliefs

Since players do not know other players' types,
they form beliefs.

Example:
Player 2 believes

$$P(\text{High Cost}) = 0.4$$

$$P(\text{Low Cost}) = 0.6$$

These probabilities are called

$$\boxed{\textbf{Beliefs}}$$

Prior Beliefs

Before the game begins,
all players know the probability distribution chosen by Nature.

This distribution is called the

$$\boxed{\textbf{Prior Distribution}}$$

Example

$$P(H) = 0.4, \qquad P(L) = 0.6.$$

Players use these probabilities to calculate expected payoffs.

Definition of a Bayesian Game

A Bayesian game consists of

1. Players
2. Strategy sets
3. Type sets
4. Prior probability distribution
5. Payoff functions

Strategies now depend on the player's type.

$$s_i : T_i \to A_i$$

Decision Making Under Uncertainty

In a Bayesian game, a player does not know the exact type of the opponent.

Instead, decisions are made using

$$\boxed{\text{Expected Payoff}}$$

The idea is simple:

Expected Payoff

Multiply each possible payoff by its probability and add them together.

This is the same principle used in probability and expected utility theory.

Example: Expected Payoff

Suppose Player 2 believes that Player 1 is

$$\begin{cases} \text{High Cost} & 40\% \\ \text{Low Cost} & 60\% \end{cases}$$

If Player 2 enters,
the payoffs are

$$5 \quad \text{(High Cost)}$$

and

$$1 \quad \text{(Low Cost)}$$

Expected payoff
$0.4(5)+0.6(1)=2.6$.

Strategies in Bayesian Games

Unlike ordinary games,
a player's strategy depends on their type.

Example

Type	Action
High Cost	Exit
Low Cost	Enter

A strategy specifies an action for every possible type.

Bayesian Strategy

A Bayesian strategy is a function

$$s_i : T_i \to A_i$$

This means

$$\boxed{\text{Type} \longrightarrow \text{Action}}$$

Example

$$H \to \text{Exit}$$
$$L \to \text{Enter}$$

Different types may choose different actions.

Bayesian Nash Equilibrium

Definition

A Bayesian Nash Equilibrium is a strategy profile in which every player's strategy maximizes their expected payoff given

- their own type,
- their beliefs about other players' types,
- the strategies of the other players.

Intuition Behind Bayesian Nash Equilibrium

Each player asks

"Given what I know,
and what I believe about everyone else,
what is my best action?"

If every player is answering this question optimally,
the resulting strategy profile is a

$\boxed{\text{Bayesian Nash Equilibrium.}}$

Nash Equilibrium vs Bayesian Nash Equilibrium

Nash Equilibrium	Bayesian Nash Equilibrium
Complete information	Incomplete information
Known payoffs	Unknown types
Strategy	Strategy for every type
Best response	Expected best response

Example: Entry Game

A new company considers entering a market.

The existing firm can be

$$\boxed{\text{Strong}}$$

or

$$\boxed{\text{Weak}}.$$

Only the existing firm knows its true strength.
The entrant only knows

$$P(\text{Strong}) = 0.7, \qquad P(\text{Weak}) = 0.3.$$

Strategies of the Incumbent

The incumbent chooses its action based on its type.

Type	Action
Strong	Fight
Weak	Accommodate

The entrant must decide whether to enter using only its beliefs.

How Do We Solve a Bayesian Game?

To solve a Bayesian game,

1. Identify the players.
2. Identify the possible types.
3. Specify the prior probabilities.
4. Write a strategy for every type.
5. Compute expected payoffs.
6. Find mutual best responses.

The resulting strategy profile is the

$$\boxed{\text{Bayesian Nash Equilibrium.}}$$

Harsanyi Transformation

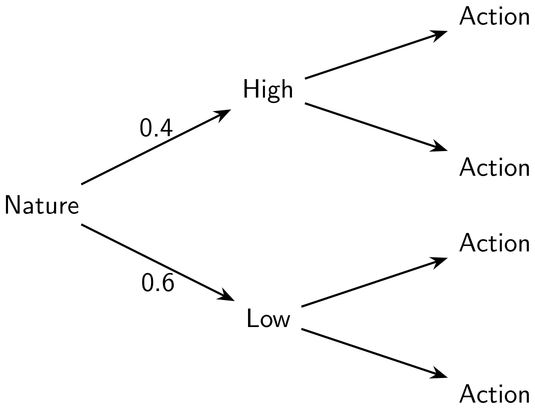
John Harsanyi showed that every Bayesian game can be represented as an extensive-form game.

The idea is simple:

1. Nature first chooses each player's type.
2. Players observe their own type.
3. Players choose actions.
4. Payoffs are determined.

This converts a game of incomplete information into one with imperfect information.

Harsanyi Transformation



Nature chooses the type before any strategic decision is made.

Application: Entry Deterrence

An incumbent firm may be

Strong

or

Weak.

A new firm must decide whether to

- Enter
- Stay Out

The entrant does not know the incumbent's true type.

Information Structure

Player	Information
Incumbent	Knows its type
Entrant	Knows only probabilities
Nature	Chooses the type

The entrant makes decisions based on beliefs rather than certainty.

Why Can't We Use Nash Equilibrium?

Ordinary Nash Equilibrium assumes

- everyone knows the payoff matrix,
- everyone knows every player's incentives.

In Bayesian games,
players do not know the exact game being played.

They only know

$$\boxed{\text{Probabilities of different games.}}$$

Example: Job Market Signaling

A worker may be

High Ability

or

Low Ability.

The employer cannot directly observe ability.

Instead, the employer observes

$\boxed{\text{Education}}$

and decides whether to hire.

Example: Poker

Poker is a classic Bayesian game.

Each player knows

- their own cards,
- betting history.

Players do not know

- opponents' cards,
- opponents' intentions.

Optimal decisions are based on beliefs and probabilities.

Example: Auctions

Each bidder has a private valuation.

Nature chooses

$$v_i$$

for every bidder.

Each bidder knows
only their own valuation.

The challenge is to determine the optimal bid.

Example 1: Entry Deterrence Game

Consider a market with two firms.

Players

- Incumbent Firm
- Potential Entrant

The incumbent may be one of two types.

$$T = \{\text{Strong}, \text{Weak}\}$$

Nature chooses

$$P(S) = 0.7, \qquad P(W) = 0.3.$$

Only the incumbent observes its type.

Available Actions

The entrant chooses

$\boxed{\text{Enter}}$ $\boxed{\text{Stay Out}}$

If entry occurs,
the incumbent chooses

$\boxed{\text{Fight}}$ $\boxed{\text{Accommodate}}$

Question:
Should the entrant enter the market?

Payoff Structure

Suppose the entrant receives

Incumbent Type	Fight	Accommodate
Strong	1	3
Weak	4	6

If the entrant stays out,
its payoff is

$$0.$$

Step 1: Form Beliefs

The entrant does not know the incumbent's type.

Instead,
the entrant believes

$$P(S) = 0.7, \qquad P(W) = 0.3.$$

These beliefs are used to calculate the expected payoff of entering.

Step 2: Expected Payoff

Suppose

- Strong firms always Fight.
- Weak firms always Accommodate.

Expected payoff from entering

$$0.7(1) + 0.3(6)$$

$$= 2.5.$$

Expected payoff from staying out

$$= 0.$$

Step 3: Best Response

Since

$$2.5 > 0,$$

the entrant's best response is

$$\boxed{\text{Enter.}}$$

The incumbent's strategy is

$$\boxed{S \rightarrow \mathit{Fight}}$$

$$\boxed{W \rightarrow \mathit{Accommodate}}$$

Bayesian Nash Equilibrium

The equilibrium strategy profile is

$$\boxed{(\text{Enter}, S \to \textit{Fight}, W \to \textit{Accommodate})}$$

Notice that

- the entrant chooses one action,
- the incumbent chooses an action for each possible type.

This is precisely a Bayesian strategy.

General Procedure

Whenever you solve a Bayesian Game, follow these steps.

Identify Types
↓
Nature Assigns Types
↓
Form Beliefs
↓
Compute Expected Payoffs
↓
Find Best Responses
↓
Bayesian Nash Equilibrium

Bayesian Nash Equilibrium

Seller's strategy

Type	Action
Good	*Sell*
Lemon	*Sell*

Buyer's strategy

$\boxed{\text{Buy}}$

given the prior beliefs.